

LECTURE 3

Database Languages and Functions

CSC270 · Database Systems

COMSATS University Islamabad, Lahore Campus

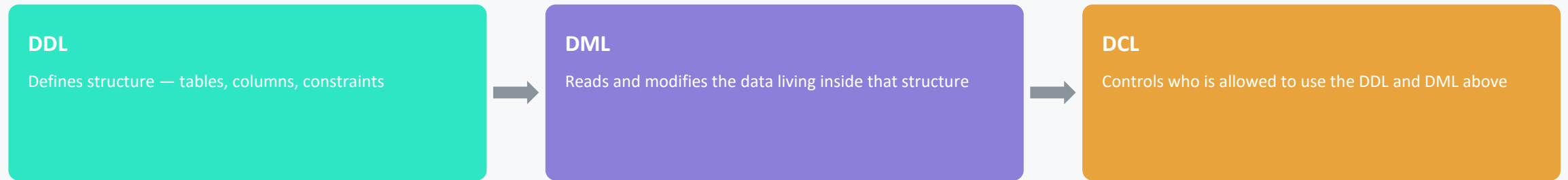
In This Lecture

- Why database languages are split into three sub-languages by purpose
- Data Definition Language (DDL): defining structure
- Data Manipulation Language (DML): working with data, procedural vs. declarative
- Data Control Language (DCL): security and access control
- The complete list of core DBMS functions, expanded from Lecture 2
- Database utilities and the day-to-day tasks of database administration

Why Split the Language by Purpose?

Talking to a DBMS involves three genuinely different kinds of statements: what the data looks like, working with the data itself, and who is allowed to do either.

The three sub-languages of a database language



One language, three jobs. In a modern relational DBMS, DDL, DML, and DCL are usually all part of the same language — SQL — rather than three separate languages learned one at a time. The split here is by purpose, so each part can be reasoned about on its own; SQL itself is covered later in this course, once the relational model is established.

Data Definition Language (DDL)

DDL defines, alters, and removes the database's structure — its schema — rather than the data inside it. For a library system, DDL creates the Book and Member relations, and later adds a column Book didn't originally have.

Book (isbn, title, author, available)

isbn	title	author	available
978-0132350884	Clean Code	Robert C. Martin	true
978-0201633610	Design Patterns	Gamma, Helm, Johnson, Vlissides	false



Create

A new table, view, index, or entire database



Alter

Add, remove, or modify a column; add a constraint



Drop

Remove a table, view, or index — and typically its data

Every DDL statement, when executed, updates the system catalog — which is precisely why DDL changes are felt everywhere at once: every part of the DBMS that consults the catalog immediately sees the new structure.

Data Manipulation Language (DML)

Where DDL shapes the container, DML works with what's inside it: retrieving existing data and inserting, updating, or deleting rows.

Retrieval

Asking questions of existing data (“which books by Robert C. Martin are currently available?”) without changing anything

Update

Inserting new data, modifying existing data, or deleting data

Procedural vs. declarative DML — a distinction worth remembering early

Procedural DML

Specify what data is needed AND how to get it, step by step — what relational algebra does (covered later)

Declarative DML

Specify only what data is needed; the optimizer decides how — what SQL and relational calculus do

A single logical action — “a member returns Clean Code” — typically involves more than one DML statement working together: marking the book available, and removing the corresponding loan record. Lecture 29's transaction concept guarantees such groups either all succeed or none do.

Data Control & Security (DCL)

DCL governs permissions: who may run which DDL and DML statements against which parts of the database. Without it, every user with any access would have unrestricted access to all of it.

- Grant — give a specific privilege (read, insert, update, delete, or the right to grant further privileges) to a named user or role
- Revoke — take back a previously granted privilege

DCL in the library system — who can do what to Book

Front-desk clerk

Granted: SELECT, UPDATE (availability only)

Cataloging librarian

Granted: SELECT, INSERT, UPDATE, DELETE

Library patron (self-service)

Granted: SELECT only

DCL is one concrete mechanism behind the “authorization services” function named in Lecture 2 — the DBA uses it to translate an organization's security policy into rules the DBMS actually enforces on every request.

Database Utilities

Lecture 2's function list rounds out with the operational side. Utilities are supporting programs, usually bundled with the DBMS, that help administer the database without being part of the core query engine:

Utility	Purpose	Library-system example
Loading	Bulk-importing data from an external source	Importing 10,000 records from the old system
Unloading / export	Extracting data out to a file	Exporting overdue-loan records for a report
Backup	Making a recoverable copy of the database	Nightly backup of the entire library database
Reorganization	Restructuring physical storage for performance	Rebuilding a fragmented index on Book.isbn
Performance monitoring	Tracking query speed and resource use	Noticing “search by author” has gotten slow
Statistics analysis	Gathering data distribution info for the optimizer	Recording 40% of loans are computer-science books

Database Administration Functions

Database administration is the ongoing, human side of keeping a database healthy — the day-to-day work of the DBA role:

- **Managing data storage and structure** — physical storage decisions and creating DDL-defined structures
- **Enforcing security and authorization** — issuing and revoking DCL privileges as staff join, change roles, or leave
- **Monitoring performance and tuning** — watching for slow queries and adjusting indexes or configuration
- **Managing backup and recovery** — scheduling backups and rehearsing recovery before a real failure
- **Maintaining data integrity** — reviewing that constraints still reflect the organization's rules
- **Liaising with users** — translating user and developer needs into database changes

A common mix-up: utilities vs. administration. Utilities are tools; administration is the ongoing responsibility that uses those tools. A backup utility can exist for years without protecting anyone if no administrator schedules it to run.

Key Takeaways

- Database languages split by purpose into DDL (define structure), DML (work with data), and DCL (control access) — usually unified into one language such as SQL
- DDL creates, alters, and drops schema objects; every DDL statement updates the system catalog immediately
- DML retrieves and updates data; it can be procedural (specify how) or declarative (specify only what) — modern SQL is predominantly declarative
- DCL grants and revokes privileges, turning security policy into rules the DBMS enforces on every request
- Database utilities (load, backup, reorganize, monitor) are the tools; database administration is the DBA's ongoing responsibility that relies on them day to day